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Tugas Jobsheet 15

Customer.java:

```
package Jobsheet15;
```

```
public class Customer {
```

```
    public int id;
```

```
    public String name;
```

```
    Customer(){
```

```
}
```

```
    Customer(int id, String name){
```

```
        this.id = id;
```

```
        this.name = name;
```

```
}
```

```
    public String toString(){
```

```
        return " ID: " + this.id + " Nama: " + this.name;
```

```
}
```

```
}
```

Book.java:

```
package Jobsheet15;
```

```
public class Book {
```

```
String isbn;
```

```
String title;
```

```
Book(){
```

```
}
```

```
Book(String isbn, String title){
```

```
    this.isbn = isbn;
```

```
    this.title = title;
```

```
}
```

```
public String toString(){
```

```
    return "ISBN: " + this.isbn + " Title: " + this.title;
```

```
}
```

```
}
```

DemoArrayList.java:

```
package Jobsheet15;
```

```
import java.util.ArrayList;
```

```
import java.util.Collections;
```

```
public class DemoArrayList {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<Customer> customer = new ArrayList<>(2);
```

```
        Customer customer1 = new Customer(1, "Zakia");
```

```
        Customer customer2 = new Customer(2, "Budi");
```

```
customer.add(customer1);
```

```
customer.add(customer2);
```

```
customer.add(new Customer(4, "Cica"));
```

```
customer.add(2, new Customer(100, "Rosa"));
```

```
System.out.println(customer.indexOf(customer2));
```

```
Customer customers = customer.get(1);
```

```
System.out.println(customers.name);
```

```
customers.name = "Budi Utomo";
```

```
for (Customer c : customer) {
```

```
    System.out.println(c.toString());
```

```
}
```

```
ArrayList<Customer> newCustomers = new ArrayList<>();
```

```
newCustomers.add(new Customer(201, "Della"));
```

```
newCustomers.add(new Customer(202, "Victor"));
```

```
newCustomers.add(new Customer(203, "Sarah"));
```

```
customer.addAll(newCustomers);
```

```
for (Customer c : customer) {
```

```
    System.out.println(c.toString());
```

```
}
```

```
System.out.println(customer);
```

```
ArrayList<String> daftarSiswa = new ArrayList<>();
```

```
daftarSiswa.add("Zainab");
```

```
daftarSiswa.add("Andi");
```

```
daftarSiswa.add("Rara");
```

```
Collections.sort(daftarSiswa);
```

```
System.out.println(daftarSiswa);
```

```
customer.sort((c1, c2)->c1.name.compareTo(c2.name));
```

```
System.out.println(customer);
```

```
}
```

```
}
```

StackDemo.java:

```
package Jobsheet15;
```

```
import java.util.Stack;
```

```
public class StackDemo {
```

```
    public static void main(String[] args) {
```

```
        Book book1 = new Book("1234", "Dasar Pemrograman");
```

```
        Book book2 = new Book("7145", "Hafalah Shalat Delisa");
```

```
        Book book3 = new Book("3562", "Muhammad Al-Fatih");
```

```
        Stack<Book> books = new Stack<>();
```

```
        books.add(book1);
```

```
        books.add(book2);
```

```
books.add(book3);
```

```
Book temp = books.peek();
```

```
if (temp != null) {  
    System.out.println(temp.toString());  
}
```

```
Book temp2 = books.pop();
```

```
if (temp2 != null) {  
    System.out.println(temp2.toString());  
}
```

```
for (Book book : books) {  
    System.out.println(book.toString());  
}
```

```
System.out.println(books);
```

```
System.out.println(books.search(book1));  
}  
}
```

TreeSetDemo.java:

```
package Jobsheet15;
```

```
import java.util.TreeSet;
```

```
public class TreeSetDemo {
```

```

public static void main(String[] args) {
    TreeSet<String> fruits = new TreeSet<>();

    fruits.add("Mangga");
    fruits.add("Apel");
    fruits.add("Jeruk");
    fruits.add("Jambu");

    for (String temp : fruits) {
        System.out.println(temp);
    }

    System.out.println("First: " + fruits.first());
    System.out.println("Last: " + fruits.last());

    fruits.remove("Jeruk");
    System.out.println("Setelah remove: " + fruits);

    fruits.pollFirst();
    System.out.println("Setelah poll first: " + fruits);

    fruits.pollLast();
    System.out.println("Setelah poll last: " + fruits);
}
}

```

Hasil:

```

1
Budi
ID: 1 Nama: Zakia
ID: 2 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
ID: 1 Nama: Zakia
ID: 2 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
ID: 201 Nama: Della
ID: 202 Nama: Victor
ID: 203 Nama: Sarah
[ ID: 1 Nama: Zakia, ID: 2 Nama: Budi Utomo, ID: 100 Nama: Rosa, ID: 4 Nama: Cica, ID: 201 Nama: Della, ID: 202 Nama: Victor, ID: 203 Nama: Sarah]
[Andi, Rara, Zainab]
[ ID: 2 Nama: Budi Utomo, ID: 4 Nama: Cica, ID: 201 Nama: Della, ID: 100 Nama: Rosa, ID: 203 Nama: Sarah, ID: 202 Nama: Victor, ID: 1 Nama: Zakia]

```

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemrograman
ISBN: 7145 Title: Hafalah Shalat Delisa
[ISBN: 1234 Title: Dasar Pemrograman, ISBN: 7145 Title: Hafalah Shalat Delisa]
2

```

```

Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah remove: [Apel, Jambu, Mangga]
Setelah poll first: [Jambu, Mangga]
Setelah poll last: [Jambu]

```

Praktikum 13.3:

1. Apakah object dapat ditambahkan meskipun melebihi kapasitas?

Jawaban: Bisa

2. Di awal, di tengah, atau di akhir collection?

Jawaban: Di akhir collection

3. Index pada ArrayList dimulai dari 0 atau 1?

Jawaban: Index dimulai dari 0

4. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

Jawaban: ArrayList tetap dapat diinstansiasi meskipun tidak ditentukan size awal

Praktikum 13.4:

1. Mengapa perlu ada pengecekan (temp != null)?

Jawaban: Untuk melakukan pengecekan dimana apakah temp merupakan null atau tidak, berarti apakah temp memiliki data yang kosong atau tidak

2. Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

Jawaban: Dengan menggunakan sout maka akan mendapatkan index dari data yang ingin dicari, ini menentukan Lokasi data yang di search tersebut

Praktikum 13.5:

1. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

Jawaban: Karena pada dasarnya TreeSet secara otomatis melakukan sorting nama dari A hingga Z

2. Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

Jawaban: first() menampilkan data terdepan, sedangkan last() menampilkan data yang paling belakang. Remove() melakukan penghapusan data sesuai yang di inginkan, pollFirst() menghapus data terdepan, dan pollLast() menghapus data yang paling belakang